

Extreme tree growth responses...

Victor, Ben, Mike, Lizzie

Tree-ring data offer a unique opportunity to learn how trees grow in a non-stationary climate with increasing extremes. However, disentangling tree responses to long-term trends and extreme events remains challenging due to the complex interactions between multiple climatic drivers and size-dependent processes. We developed a new modeling framework that integrates individual-level growth trends with species-specific climate sensitivities. Our model is specifically designed to capture extreme growth responses across trees, species and ecosystems. We applied it to tree-ring records from Western North America, from 1900 to today. We identify the significant drought period of the 200X-200Y as causing exceptional reductions in tree growth, well beyond those predicted alone due to temperature, winter precipitation (a proxy of soil moisture during growing season) and vapor pressure deficit. The largest declines occurred in XX species and YY regions. These results demonstrate that tree growth is a critical indicator of drought, and suggest that many current models may underestimate growth declines associated with extreme drought events.

Temperature, vapor pressure deficit, and soil moisture with anthropogenic climate change are causing new extremes for trees. Understanding their impact on tree growth, however, remains challenging given limited large-scale soil moisture data and the complex interaction between multiple climate drivers and size-dependent tree growth. We developed a new modeling framework that integrates individual-level growth trends with species-specific climate sensitivities. Our model is specifically designed to capture extreme growth responses across trees, species and ecosystems. Using new WLDAS data to get soil moisture, we apply the model to 1.6 million observations of tree-ring width across Western North America. We identify the significant drought period of the 200X-200Y as causing exceptional reductions in tree growth, well beyond those predicted alone due to temperature, vapor pressure deficit and soil moisture. The largest declines occurred in XX species and YY regions. These results demonstrate that tree growth is a critical indicator of drought, and suggest that many current models may underestimate growth declines associated with extreme drought events.